

AWS checklist and maintenance guide

Station name	M.I.T
People present	RSF, SIBA instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency
Date	10/8-2015
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %: 0

Notes:

Station in very bad condition. Covered in snow up to loggerbox
 Boom bent. Loose wire indicating broken wire.
 Tried to dig down, but hit ice before we could see what the problem was. Station taken down.

Visit again 13/8-2015

Set up old TAS-U station

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	15 (+/-)
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	(Y) / N
Name of logger program	
Photo or screen dump of values in fast scan mode	(Y) / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	(Y) / N
Mast tilt in boom direction (+ if radiometer tilting down)	(+20, 0) + 7 °
Mast tilt across boom looking from radiometer (- if clockwise)	+ 2 °
Boom direction relative to north	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	(Y) / N
Radiometer aligned with mast and inclinometer? If not, measure.	(Y) / N
Temperature / humidity height (bottom of casing – surface)	1.65 cm
Sonic ranger on AWS height (membrane – surface)	1.65 cm
Wind sensor height (center of propeller – surface)	2.11 cm
Sonic ranger on stakes height (membrane – surface)	?
Free length of stake – outer, holding boom with sonic ranger	?
Free length of stake – middle	?
Free length of stake – outer, holding boom without sonic ranger	?
Free length of ablation hose on/above the ice surface	?
Alt: depth of ablation hose using markings on hose	?
Height of ablation hose fluid level above the ice surface	?
Vertical difference in surface height (at station – at hose) (approx)	?
Length of thermistor string on surface from surface marking	?
Height of surface irregularities	1 cm
Stake assembly still standing?	Y / (N) / more or less
Apparent damage to sensors? If yes, which?	Y / N

boom bent
radiometer
tilting down

same as
last year

Notes:

Height of old stake assembly stakes

Down stream stake: 150

Up — 165

Snow depth: 125 Down stream

120 Up stream

Avg. snow depth
around station

1: 1.33

2: 1.23

3: 0.93

4: 0.93

5: 0.96

Station taken down.

MIT

13/8

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer		080164	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer		0008052	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna		GEVS12	
Wind sensor			- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly		6/2011	
T/hum assembly casing		5355025177	- Fan spins and sounds OK: Y/N
T/hum assembly bracket			
Temperature probe in T/hum assembly		MP1001 6079402	
HygroClip in T/hum assembly			- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS		1175	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel		01030306 2251887	- Output OK: Y/N - Clean: Y/N
GPS antenna		030110034	- Stuck to top of logger enclosure: Y/N
Data logger		AWS 02/2009	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader		4442	- LED green or flashing orange (active): Y/N
Multiplexer		E3201	- Clicking as it should: Y/N
Iridium modem IMEI number		1218580	- Transmissions tested: Y/N - Problems: Y/N
Barometer		3814092	
Sonic ranger on stakes	5110	2843	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string		X03	- Old string left in ice: Y/N/cut
Ablation hose assembly		old MIT	- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	Y / N
Name of logger program	Promise 2015e.
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	3 °
Mast tilt across boom looking from radiometer (- if clockwise)	-1 °
Boom direction relative to north (radiometer should point to true south)	° true/magn
Box on wind sensor exactly towards mast? If not, change.	Y / N
Radiometer aligned with mast and inclinometer? If not, align.	Y / N
Temperature / humidity height (bottom of casing – surface)	295 cm
Sonic ranger on AWS height (membrane – surface)	274 cm
Wind sensor height (center of propeller – surface)	312 cm
Sonic ranger on stakes height (membrane – surface)	189 cm
Free length of stake – outer, holding boom with sonic ranger	225 cm
Free length of stake – middle	212 cm
Free length of stake – outer, holding boom without sonic ranger	228 cm
Free length of ablation hose on/above the ice surface	? cm
Alt: depth of ablation hose using markings on hose	? cm
Height of ablation hose fluid level above the ice surface	80 cm
Vertical difference in surface height (at station – at hose) (approx)	? cm
Length of thermistor string on surface from surface marking	12 cm
Height of surface irregularities	10 cm

snow surface

Notes:

